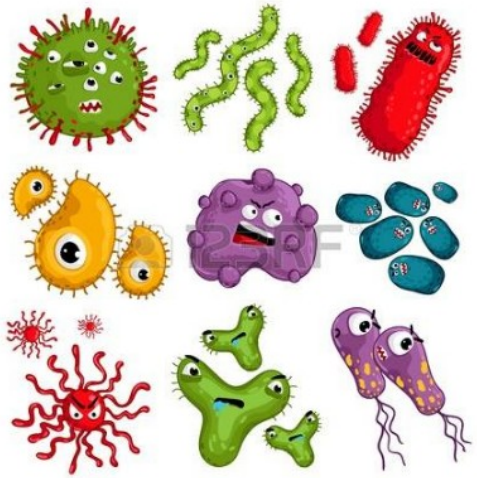
Immune System:
Pathogens, External Barriers, Protective Reflexes



The Body can Protect Itself From Infection

Vocabulary

- *Pathogens: disease causing organisms*
  - Bacteria
  - Viruses
  - Protozoa, fungi, and others

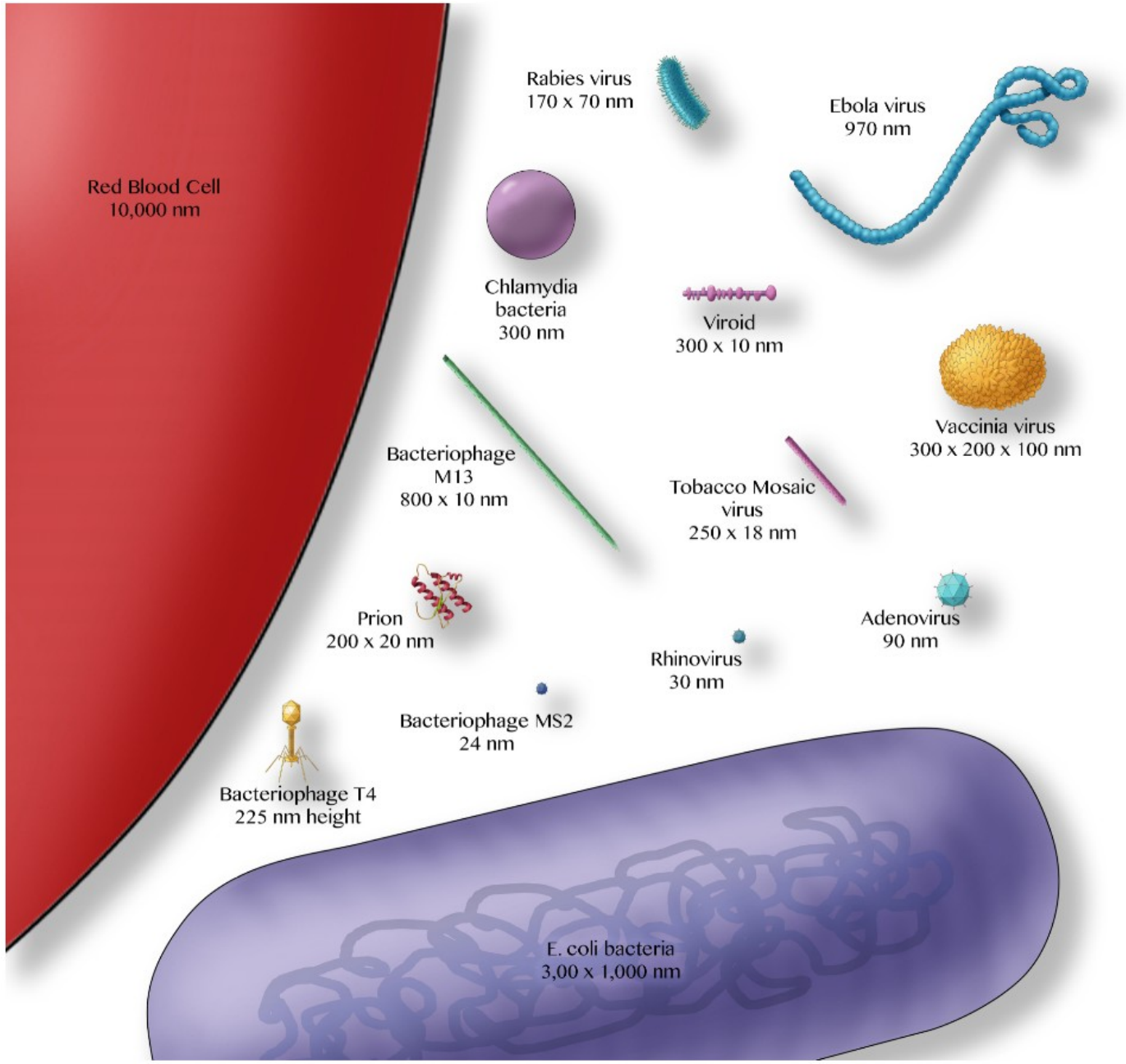


- *Infectious disease: diseases caused by pathogens*
  - Eg: A skin infection after injury is infectious, but doesn't pass on to other people. Tetanus is infectious, and can be acquired from a penetrating skin injury as the bacteria lives in soil, but is not communicable. If you have tetanus, you can't directly pass it on to anyone else.

- *Communicable/contagious disease: infectious disease that can be passed from one person to another*
  - Eg: Covid-19 is infectious AND can be passed onto others.



Relative Pathogen Sizes



Bacteria are much smaller than an average body cell, but can still be seen with a light microscope.

Viruses are much smaller again, and can be seen with a scanning electron microscope.

Size of pathogen can be a contributing factor to disease spread. Part of the reason that rhinovirus (which causes the common cold) spreads easily is because it is tiny and many particles can travel long distances via aerosol/droplet infection

Pathogen Types: Bacteria

- Most are harmless to humans
  - Commensal/non-pathogenic
  - Found everywhere including on skin, in gut
- Some cause serious disease/death
  - These are called pathogenic bacteria
- Single cell organisms
- Can be seen using a light microscope
- Very small
- Some can be prevented via vaccination
- Some can be treated with antibiotics

Common/Well Known Bacterial Diseases	
"Strep throat", "ear infection" *Streptococcus*	Streptococcus is found naturally on skin but can infect already damaged tissue.
"Golden staph" *Staphylococcus aureus*	Staphylococcus is found naturally on skin but can infect already damaged tissue. Can be resistant to antibiotics.
Salmonella	Causes severe gastroenteritis
Tetanus *Clostridium tetani*	Causes severe, rigid paralysis and death. Vaccine and antitoxin available.
"Whooping Cough" *Bordetella pertussis*	Causes severe, prolonged coughing. Babies often die if infected. Vaccine available.
Syphilis *Treponema pallidum*	A sexually transmitted disease.
Tuberculosis *Mycobacterium tuberculosis*	Causes severe, difficult to treat lung disease, used to result in death. Now can be treated with long-term, powerful antibiotics. Vaccine available, so has been eradicated in some countries.

Learning Aim: List several pathogenic bacteria and the diseases they cause

Bacterial Cell Structure

Figure 10.2a The structure of a typical bacterial cell

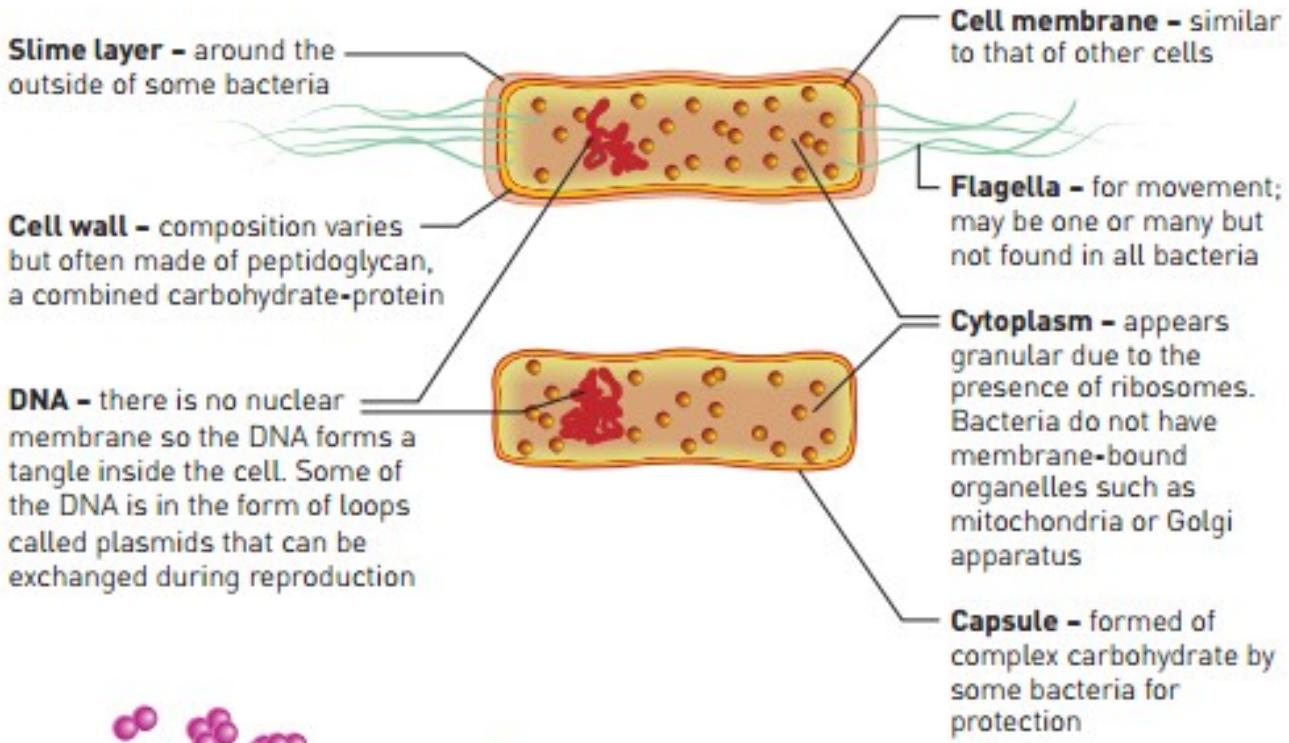


Figure 10.2b Types of bacteria, classified according to the cell shape



Cocci (singular 'coccus') are spherical cells that may occur singly, in pairs (diplococci), in clusters (staphylococci) or in chains (streptococci)

Bacilli (singular 'bacillus') with **flagella** have rod-shaped cells; many have flagella for movement

Spirilla (singular 'spirillum') have twisted cells

Vibrio are like curved rods and are often shaped like a comma

Learning Aim: Discuss the general structure of bacteria

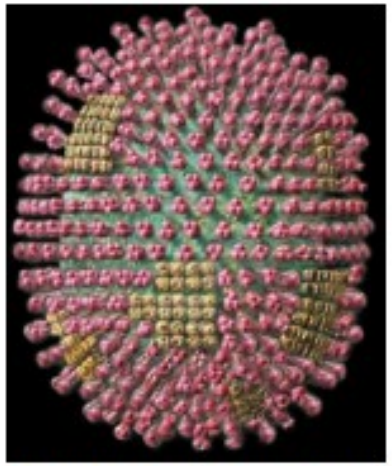
Pathogen Types: Viruses

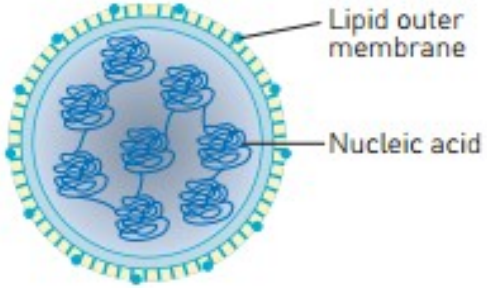
- Very small – can only be seen with electron microscope.
- Contain genetic materials eg DNA/RNA, surrounded by protein coat.
- Enter cell and use their DNA/RNA to take over cell processes so virus can reproduce.
- Not all viruses are harmful/pathogenic – some are even used in biotechnology!
- Cannot be treated with antibiotics
- Vaccines available for some deadly viral diseases

Learning Aim: List several pathogenic viruses and the diseases they cause

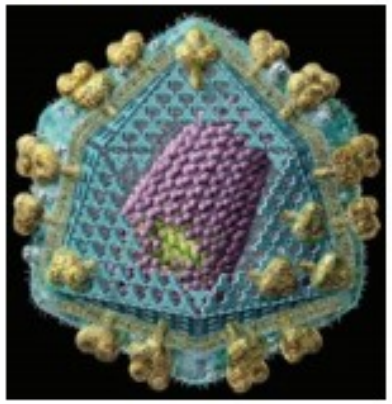
Common/Well Known Viral Diseases	
Common Cold *rhinovirus*	Causes upper respiratory tract symptoms. Self limiting, mild. Supportive treatment
Flu *Influenza virus*	Causes fever and a range of other symptoms. Moderate to severe. Supportive treatment. Preventative vaccine available.
Covid-19 *Coronavirus*	Respiratory disease with high fatality rate. Prevention via hygiene, social distancing, masking, quarantine. Preventative vaccine becoming available.
Measles *Measles virus*	Causes rash, fever, gastrointestinal illness. Usually a disease of childhood. Now rare due to vaccination.
Mumps *Mumps virus*	Causes severe, painful inflammation of the salivary glands. Now rare due to vaccination
German Measles *Rubella virus*	Causes rash, fever and other symptoms. Can cause severe birth defects or death of foetus if caught by mother. Vaccine to prevent.
Chicken Pox *Varicella virus*	Itchy rash, fever and other symptoms. Usually a disease of childhood. Becoming rarer due to vaccination.
Polio *Poliomyelitis virus*	Severe paralytic disease causing permanent disability or death. Now extremely rare due to vaccination. Eradicated in many countries.
Warts *Papilloma virus*	Several types. Some types sexually transmitted. Genital warts can cause cervical cancer. Becoming rarer due to vaccine.

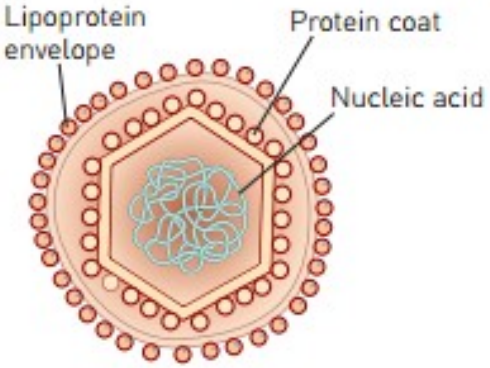
Example of how viruses use cells to replicate





Influenza virus has a lipid outer membrane. The RNA is in eight segments.

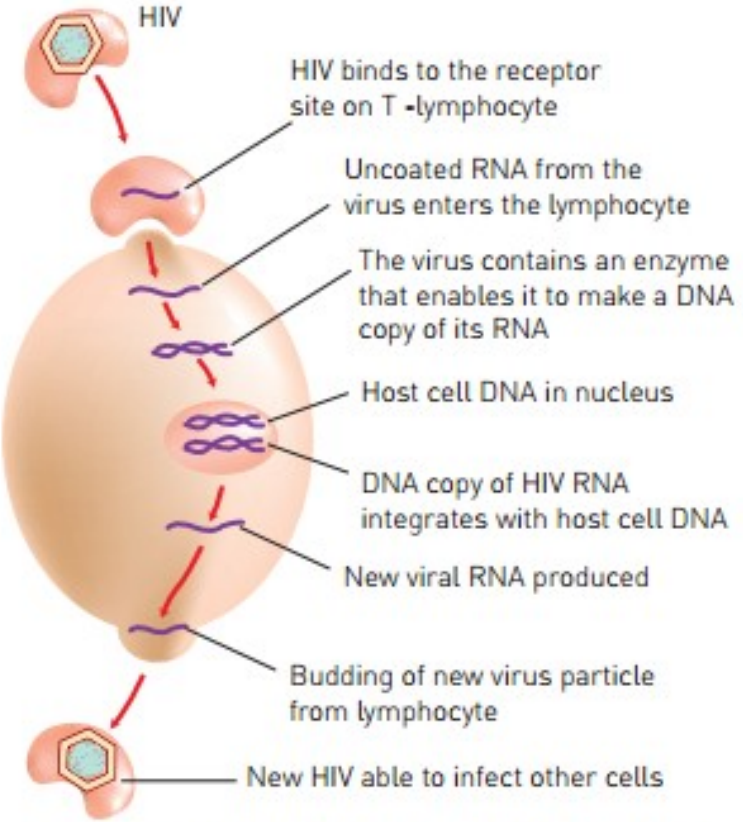




Human immunodeficiency virus (HIV) has a lipoprotein envelope with an internal protein coat.

Viruses all contain either DNA or RNA but not both. Around the nucleic acid is a protein coat, and some viruses have an additional envelope of lipid and protein molecules.

Figure 10.2c Viruses



Viruses cannot reproduce themselves. They attach to the outside of a host cell and the nucleic acid enters the cell. New viral genes are produced by the host cell, and so hundreds of new virus particles are formed.

Figure 10.2d The process of viral replication illustrated by HIV

Learning Aim: Describe the general structure of viruses and how they use cells to replicate.

A range of other pathogenic diseases not already discussed.

Table 10.1 Pathogenic organisms and some of the better-known diseases that they cause

Bacteria	Viruses	Fungi	Animal parasites
Anthrax	HIV/AIDS	Ringworm	*Protozoans*
Botulism	Bird flu	Thrush	Amoebic dysentery
Bubonic plague	Chickenpox	Tinea	Amoebic meningitis
Chlamydia	Cold sores (herpes)		Malaria
Cholera	Colds		Sleeping sickness
Dental caries (tooth decay)	Encephalitis (viral)		Toxoplasmosis
Diphtheria	Ebola		*Platyhelminthes (flatworms)*
Gastroenteritis	Genital herpes		Blood flukes
Gonorrhoea	Glandular fever		Hydatids
Impetigo (school sores)	Hepatitis A, B, C, D, E and G		Liver flukes
Legionnaire's disease	Influenza		Tapeworms
Leprosy	Measles		*Nematodes (round worms)*
Meningitis (bacterial)	Meningitis (viral)		Hookworms
Peptic ulcers	MERS (Middle East respiratory syndrome)		Roundworms
Pneumonia	Mumps		Threadworms
Scarlet fever	Poliomyelitis		*Arthropods*
Syphilis	Rabies		Lice
Tetanus	Ross River virus		Scabies (mites)
Trachoma	Rubella		Ticks
Tuberculosis	SARS (severe acute respiratory syndrome)		
Typhoid	Shingles		
Whooping cough	Smallpox		
	Warts		
	Yellow fever		

Transmission of Pathogens

Different pathogens can be transmitted in different ways.

- Contact:
  - Direct: by touching an affected person and the pathogen transfers directly eg. ringworm
  - Indirect: by touching something an affected person has touched eg. Catching cold virus from a phone used by someone with a cold.

- Transfer of Body Fluids
  - Blood, semen, mucus, faeces, urine.
  - Affected person's body fluids in contact with mucus membranes or bloodstream of another person. Eg: cold sores (a type of herpesvirus) are transmitted through salivary contact.

- Droplet/Airborne Infection
  - Pathogens spread by coughing/sneezing as droplets of mucus containing the pathogens fly through the air and are inhaled or ingested. Eg. Colds, flu, COVID-19.

- Ingestion
  - Infected matter is eaten or drunk
  - Eg. Salmonella food poisoning, cholera

- Transmission by Vector
  - *Vector* = "carrier"
  - "vector" is another organism that carries the disease from one person to another
  - Eg. *Anopheles* mosquitoes are a vector for malaria
  - Rat fleas were a vector for Bubonic Plague

Note: some pathogens are very specific in how they are transmitted.

Eg: malaria can only be transmitted via the Anopheles mosquito, but the common cold can be picked up from surfaces, via droplet infection, body fluids, or direct contact.

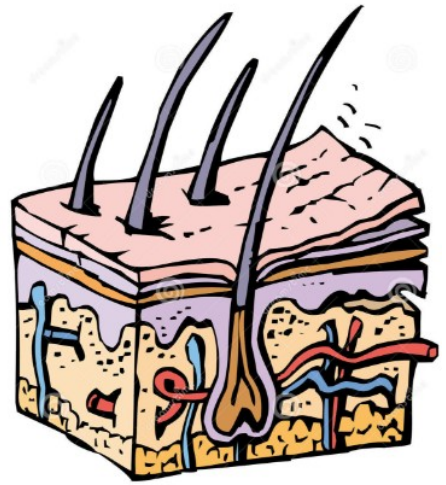
Learning Aim: Discuss modes of transmission of pathogens and give examples of diseases transmitted in each way

Defences Against Disease Overview

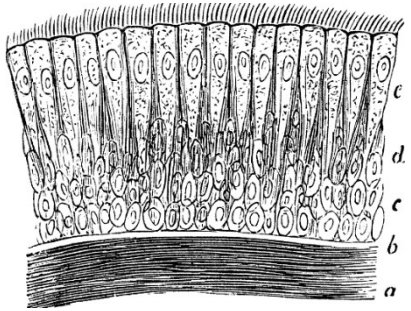
- The body has many mechanisms to protect from invading organisms that cause disease:
  - External Barriers – prevent entry of pathogens
    - skin/membranes
  - Protective Reflexes – expel pathogens from body
    - Sneezing/coughing
  - Non-specific defences – general responses to a wide range of pathogens
    - Phagocytosis/Inflammation/Fever
    - Lymphatic System
  - Specific defences/ Immune Response – responses to specific diseases.
    - Antibody mediated immunity
    - Cell Mediated Immunity

Learning Aim: List mechanisms the body uses to protect from pathogens

Defences Against Disease: External Barriers



- **Skin:**
  - Effective barrier covering outside of body
  - Stops entry of microorganisms if unbroken
  - Normal bacteria also provide protection – stop pathogenic bacteria from settling in.
  - Secretes oily *sebum* – has substances that kill some bacteria
  - Sweat can also prevent micro-organism growth
- **Mucous Membranes**
  - Eg in mouth, nose, conjunctiva, reproductive tract etc
  - Line body cavities that open to exterior
  - Secrete mucus – interferes with micro-organism entry
- **Hairs**
  - In nasal cavity, ears, eyelashes
  - Trap pathogens for processing and removal
  - *Cilia* – tiny hairs with beating motion – sweeps mucus and pathogens towards entrance
    - *Trachea*
    - *Nasal cavity*
    - *Other air passages*



Learning Aim: List and describe external barriers to disease entry.

Defences Against Disease: External Barriers

- **Acids**
  - Kill bacteria and other micro-organisms
  - Eg stomach acid, acidic vaginal secretions, sweat

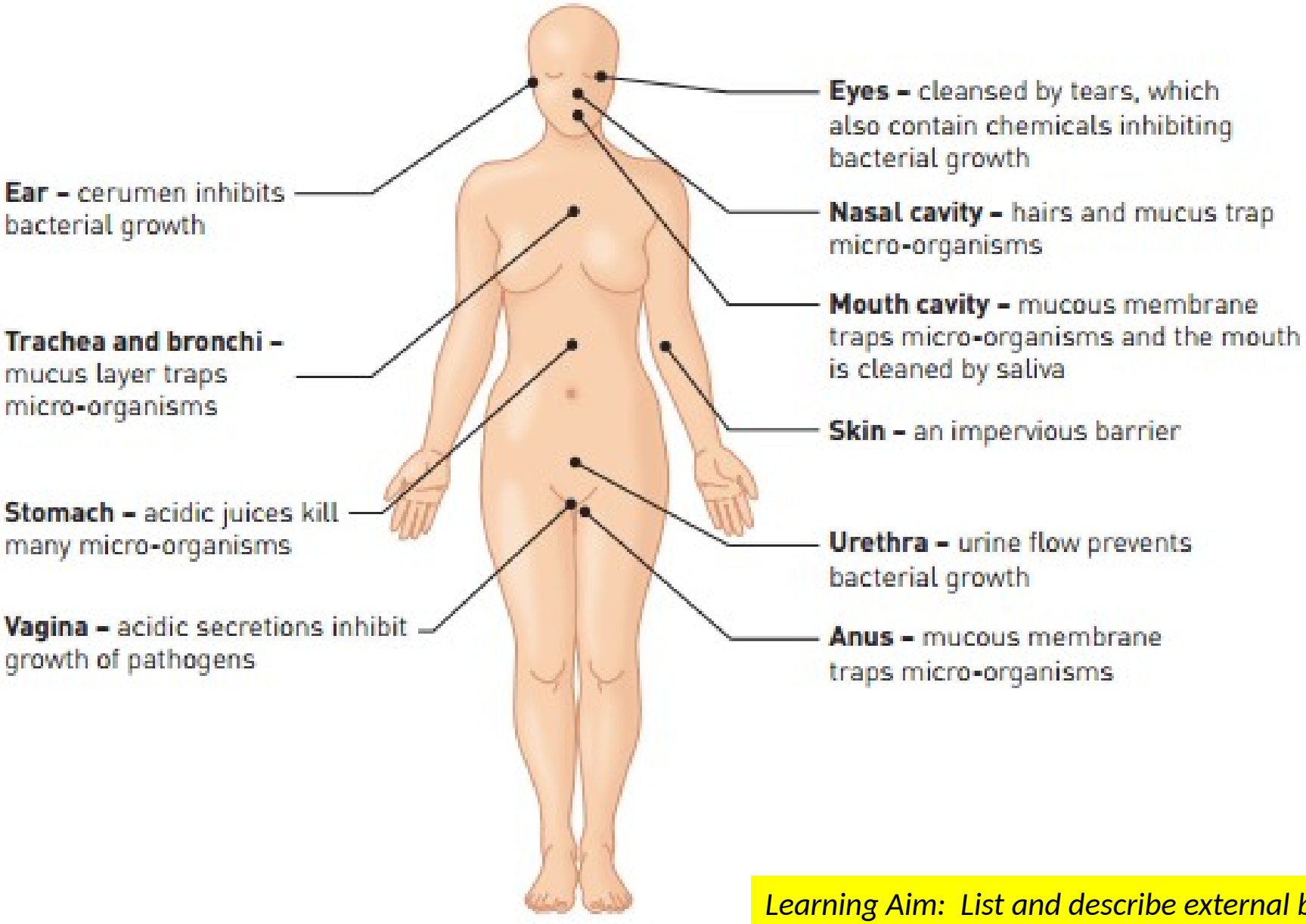
- **Lysozyme**
  - Enzyme that kills bacteria
  - Contained in tears, saliva, sweat, nasal secretions, tissue fluid

- **Cerumen**
  - Ear wax
  - Slightly acidic, contains lysozyme
  - Protects ear canal from infection

- **Flushing action of body fluids**
  - Helps keep free of pathogens

Learning Aim: List and describe external barriers to disease entry.

Defences Against Disease: External Barriers



Learning Aim: List and describe external barriers to disease entry.

Figure 10.4 The body's external defences against entry of pathogenic micro-organisms

Defences Against Disease: Protective Reflexes

- **Sneezing:**
  - Triggered by irritation of nasal cavity
  - Expels dust, dirt, mucus, and pathogens from mouth (not as much from the nose, despite what you think you know!)
- **Coughing:**
  - Irritation of lower respiratory tract
  - Air forced from lungs, taking mucus and pathogens up trachea and out
- **Vomiting:**
  - Contraction of muscles of abdomen and diaphragm
  - Expels pathogens from stomach
- **Diarrhoea:**
  - Irritation of intestinal wall causes fluid to accumulate in intestines, plus explosive contractions of smooth muscle in intestinal wall.



Learning Aim: List and describe protective reflexes that assist in defence against disease.

Assisting the Body's External Defences

- Disease can be prevented by assisting the body in preventing pathogen entry.
- Examples include:
  - Wearing a mask to prevent spread of respiratory disease when you are sick
  - Maintaining environmental hygiene – wiping surfaces, regular cleaning
  - Wearing condoms to prevent sexually transmitted disease
  - Mosquito nets to prevent mosquito-borne diseases eg malaria
  - Frequent hand washing especially:
    - Before and after food preparation
    - After toilet
    - After using a tissue or touching nasal area or removing a mask
    - After coughing or sneezing
  - Sneezing and coughing into elbow.
  - Avoiding contact with people who are sick.
  - Self-isolating when unwell.





Learning Aim: Discuss behaviours and equipment that can assist the body's external defences.